



COMPONENTS

BLDC motor driver

The new highly integrated Grade 0 brushless DC (BLDC) motor driver DRV3255-Q1 helps automobile manufacturers produce mild hybrid electric vehicles (MHEVs) with minimised emissions from a vehicle's internal combustion engine. It helps designers shrink their motor system



size by as much as 30 percent while providing a high gate-drive current for increased protection and output power. At the same time, board space and bill of materials are reduced. The DRV3255-Q1 was designed using TI's certified functional safety hardware development process and includes built-in safety mechanisms and documentation such as failure modes, effects and diagnostic analysis and a functional safety manual.

Texas Instruments

www.ti.com

Energy-efficient microcontroller

Silicon Labs has launched the EFM32PG22 (PG22) 32-bit microcontroller unit (MCU), a low-cost, high-performance solution with an industry-leading combination of energy



efficiency, performance, and security. With easy-to-use, cost-efficient analogue capabilities, the PG22 MCU is ideally suited for rapid development of consumer and industrial applications with demanding size constraints and low-power operation requirements. Offering 100 percent form factor and code alignment with xG22 SoCs, developers can take advantage of application sharing and a drop-in upgrade path to Bluetooth Low Energy (LE), Zigbee, or Proprietary (2.4GHz) wireless connectivity.

Silicon Labs

www.silabs.com

Smart TV chip

The MT9638, a new 4K smart TV chip, comes with an integrated high-performance AI processing unit to support cutting-edge AI-enhancement technologies such as AI super-resolution, AI picture quality, and AI voice assistants, along with variable refresh rate and motion estimation and motion compensation, so that graphics appear smoother. It also supports applications powered by AI voice recognition and AI-enabled video calls, opening up new ways for consumers to interact with their smart TV and deliver new, immersive experiences in streaming, gaming, and entertainment.



MediaTek

www.mediatek.com

Enhanced ultrasonic sensors

Offering a range of transmitter, receiver, and transceiver options with analogue outputs, the newly launched ultrasonic sensors feature distance ratings from 0.2 to 18 metres and beam angles of 75 or 80 degrees.



These are housed in compact aluminium or plastic cases with through-hole mounting styles, providing an economical solution for precise measurement in a variety of detection, distance, and proximity applications. With frequency ratings of 23 to 40kHz, peak-to-peak voltage ratings of 80 to 180V, and operating temperature from -30 degrees to +85 degrees Celsius, these ultrasonic sensors offer sound pressure levels from 105 to 115dB and sensitivity ratings from -75 to -68dBV.

CUI Devices

www.cuidevices.com

High-efficiency switching regulator

Traco Power's new TSR 1.5E series of 1.5-ampere step-down switching regulators



are modern drop-in replacements for older, inefficient step-down regulators in the industry-standard SIP-3/TO-220 footprint. This new generation of non-isolated DC-DC converters is cost-optimised to support price-sensitive and high-volume requirements. These drop-down regulators provide high-efficiency operation up to 97 percent and provide full-load operation over an ambient temperature of -40 degrees to +85 degrees Celsius without heat sinks, temperature derating, forced-air cooling, or external capacitors.

Traco Power

www.tracopower.com

Array for LiDAR

The new RDM-series silicon photomultiplier (SiPM) ArrayRDM0112A20-QFN extends the LiDAR sensor capabilities



to intelligent sensing solutions for the automotive industry and beyond. It is a monolithic 1×12 array of SiPM pixels, which enables high sensitivity to near-infrared light to achieve industry-leading 18.5% photon detection efficiency at 905nm. The high internal gain of the SiPM allows sensitivity down to the single-photon level. This results in the ability to range to greater distances even with low reflective targets.

ON Semiconductor

www.onsemi.com

Precision measurement ICs

Designers can now achieve double the battery life, robust protection, and highest accuracy for IoT, industrial, and healthcare applications with three new Analog ICs from Maxim—MAX41400 instrumentation ampli-